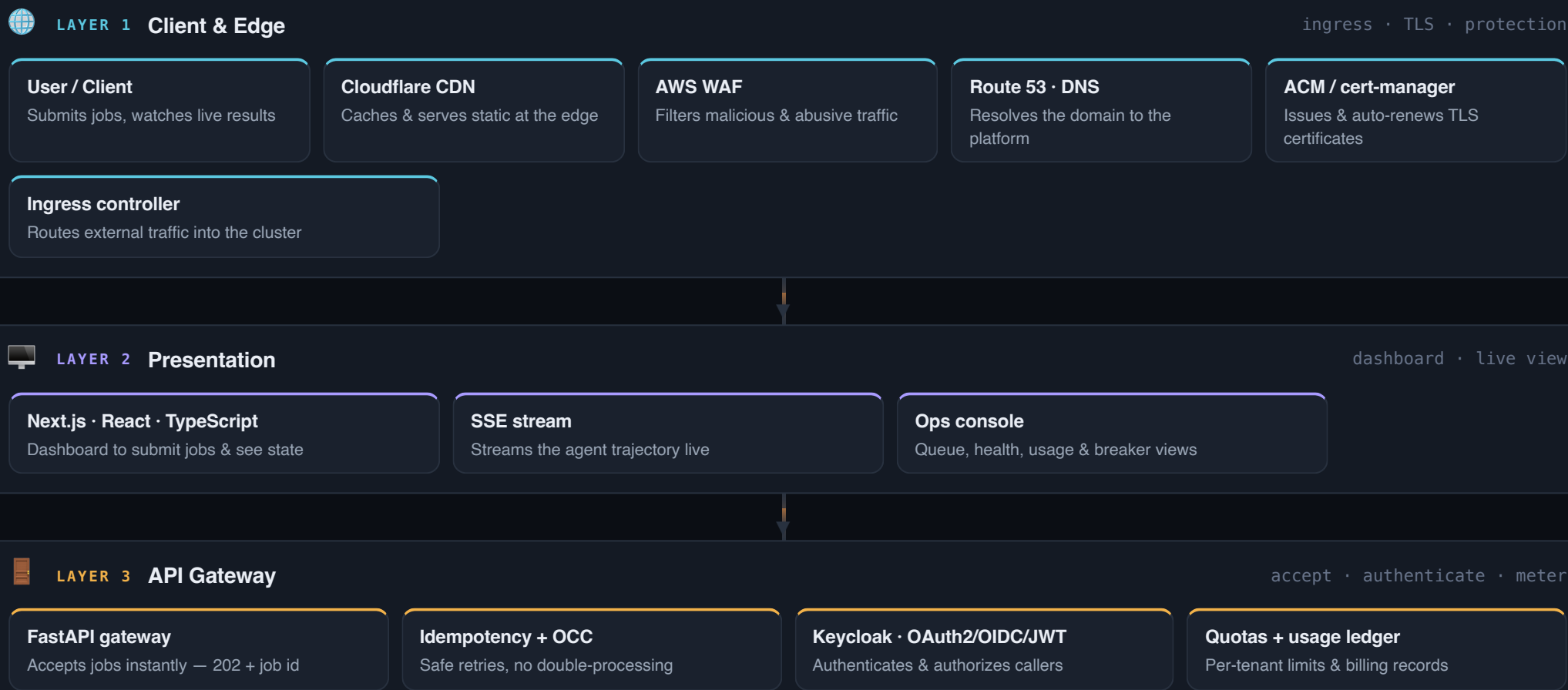


Helios — End-to-End Architecture

Cloud-Native LLM Orchestration Platform · how a request flows from edge to model to storage

A client submits a job at the **edge** → the **FastAPI gateway** accepts it instantly (202 + id) → it's queued on the **Kafka** event backbone → a **Go worker** runs the **agent loop** against **Claude**, wrapped in **resilience** (rate-limit · breaker · retry · fallback) → results stream back over **SSE** and persist across the **polyglot stores** — all running on **Kubernetes**, shipped by **GitOps**, observed end-to-end.

Edge Presentation Gateway Resilience Event bus Workers Storage Platform Delivery Observability ▼ animated line = live request flow



LAYER 4 Control plane

routing · experiments · provisioning

Provider registry & selector

Routes each job to a healthy provider

Experimentation / feature flags

A/B-tests agent configs & models

Preview-env provisioner

Custom Operator spins up tenant stacks

LAYER 5 Resilience layer

wraps every provider call

Rate limiter

Token bucket (Redis/Lua) — respects limits

Circuit breaker

Stops calling a failing provider fast

Retry + backoff + jitter

Recovers from transient errors

Fallback chain

Degrades to a backup provider

Load shedding

Drops excess load to stay alive

LAYER 6 Event backbone

durable · decoupled · replayable

Kafka / Redpanda

Durable job & event stream

Schema Registry

Versioned, evolvable event contracts

Outbox pattern

Atomic DB-write + publish

Dead-letter queue

Quarantines poison messages

Debezium · CDC

Streams DB changes to the warehouse

LAYER 7 Worker fleet · Go

the agent runtime

Go workers

Concurrent job processors (goroutines)

Agent loop

plan → act → observe reasoning cycle

Tool / function calling

Lets the agent invoke real tools

Token budget

Caps cost & length per job

RAG retrieval

Grounds answers in your own data

gRPC + Protobuf

Fast typed internal service calls

Istio mesh

mTLS + traffic shaping between services

calls the model →



LAYER 8 LLM providers

the intelligence

Anthropic Claude API

The model that powers the agents (Messages + tool use)

persist & retrieve ↕



LAYER 9 Storage plane · polyglot

right store for each shape of data

PostgreSQL

Jobs, tenants, quotas, ledger

MongoDB

Agent trajectories & payloads

Redis

Hot state, cache, rate-limit counters

pgvector / Qdrant

Embeddings for RAG search

S3 / MinIO

Artifacts & long-term archive

ClickHouse

Analytics & CQRS read model

SPANS EVERY LAYER ABOVE



CROSS-CUTTING Platform & runtime

where everything runs

Kubernetes · EKS + GKE

Runs every service on the fleet

Helm + Kustomize

Packages & configures deployments

Custom Operator + CRD

Automates tenants & preview envs

KEDA

Autoscales workers on Kafka lag

HashiCorp Vault

Dynamic secrets & PKI



CROSS-CUTTING Delivery & supply chain

how code reaches production

GitHub Actions + OIDC

CI with keyless cloud auth

Argo CD · GitOps

Declarative continuous deployment

Argo Rollouts

Canary & blue-green releases

Terraform / Crossplane

Provisions cloud infra as code

SBOM · cosign · SLSA

Signs & verifies the supply chain



CROSS-CUTTING Observability & SRE

is it healthy? is it fast?

OpenTelemetry

Distributed traces across services

Prometheus

Metrics & alerting rules

Loki

Centralized structured logs

Grafana

Dashboards for the whole system

Pyroscope

Continuous profiling of hot paths

SLOs + burn-rate alerts

Reliability targets & paging

Chaos Mesh

Fault-injection experiments

k6

Load & performance testing

Helios · Cloud-Native LLM Orchestration Platform – end-to-end architecture · every tool shown with its use case · the animated flow line traces one live request from edge to model